

CTA L Ridership Analysis Pre and Post COVID Crisis

2026-09-17

Introduction

This project is an analysis of the CTA's L (rail) ridership before and after the onset of the COVID-19 crisis. It is well known that the onset of the pandemic—"lockdowns," curfews, and restrictions in most places in the USA caused immediate and dramatic changes in transport, work, and safety habits. These disruptions were most severe during the early months of the crisis, but their effects carried onward, most being measurable today. Some examples of these effects include shifts in transit methods and popularity of remote work/hybrid scheduling. I will examine how ridership changed following the onset of the pandemic and whether the extent of ridership recovery varied according to economic (income) characteristics of the census tracts containing the CTA stations.

First, I examine CTA L ridership trends spanning from 2016 to 2024 to identify key changes in overall ridership, as well as the magnitude. Pre-pandemic monthly ridership in the year 2019 is the baseline that will be compared to the corresponding post pandemic onset monthly ridership. This accounts for seasonality in CTA ridership, to distinguish between routine variation and crisis-correlated variation in transit (rail) use.

Then, I examine ridership recovery across stations contained in areas varying in income level. CTA stations span across and serve areas with diverse economic characteristics, which the crisis may not have affected uniformly. To investigate, I combine station-level CTA ridership data with income data collected by the American Community Survey (ACS) in 2023.

This project addresses two primary questions:

1. How did CTA L ridership change, following the onset of the COVID-19 crisis?
2. What were the variations in CTA L ridership recovery according to the economic characteristics (income) of the census tract that contains the CTA station?

I address the first question through descriptive analysis by plotting ridership over time and identifying overall trends visually. The second question is addressed using station level monthly ridership, with each station categorized by income group. I use correlation analysis and linear regression to investigate the correlation between station area income and its ridership recovery. The goal is to establish existing association, not causality.

Overall Ridership Trends

1 Did CTA L Ridership change post-onset of the pandemic?

Yes, ridership changed drastically with a steep cutoff at the beginning of 2020, following the onset of the COVID crisis.

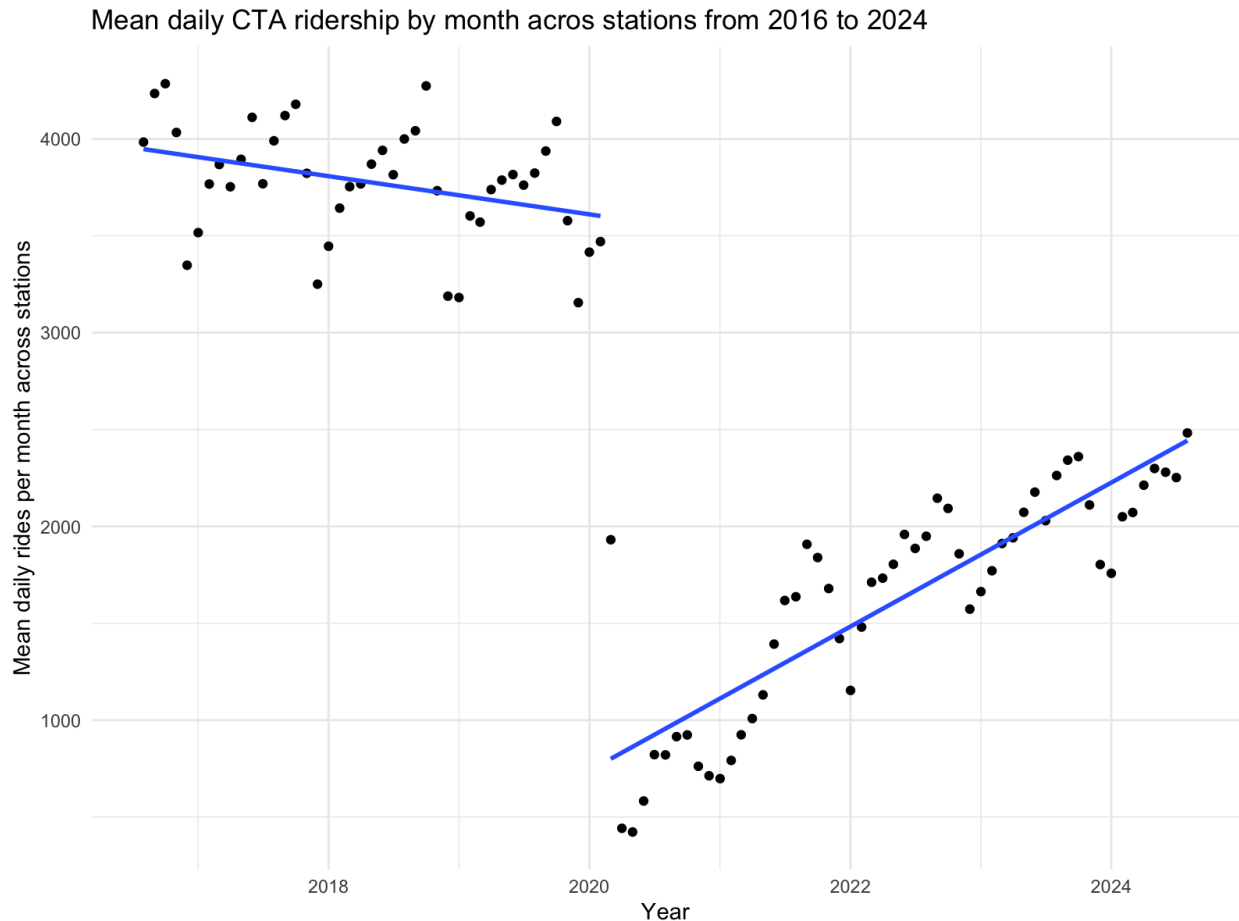


Figure 1: figure 01, Mean daily CTA ridership by month from 2016 to 2024.

In the making of this figure, daily ridership records were aggregated by month, and average ridership per day across all stations was taken. Ridership data was also split into two categories—“true” and “false,” pertaining to whether or not the record was pre or post crisis onset.

2 Did CTA L Ridership “recover” post-onset of the pandemic?

Yes, ridership experienced a gradual “recovery,” a return to its baseline (of the corresponding month in 2019) after the onset of the COVID crisis.

Q1: How did CTA L ridership change, following the onset of the COVID-19 crisis?

Ridership after the start of the crisis experienced a steep decline of daily rides per month across all stations during 2020. Between 2020-02-01 and 2020-04-01, the decline in average daily number of rides across all stations is 3,028, in total, system-wide ridership declined by 10,309,970 rides, and percent decline was 86.7%. It is also notable that prior to the cut-off, ridership had a slight, preexisting decline before the pandemic started.

A gradual, but incomplete recovery by 2024 can be observed on the plot of mean daily rides per month and year. By looking at the plot itself, it is difficult to measure the magnitude of recovery, with recovery being defined as the proportion post-crisis ridership of to its baseline pre-pandemic ridership, which leads us to the next topic.

To investigate the magnitude of ridership recovery, I created a recovery metric, proportion of baseline (“prop_of_baseline”). The *baseline* is “normal” ridership, which I reference ridership in 2019 for. Each post-crisis month is matched to the corresponding baseline month in 2019 to account for seasonality throughout the year. Recovery is defined as the ratio of mean daily ridership each month to mean daily ridership in its corresponding baseline month.

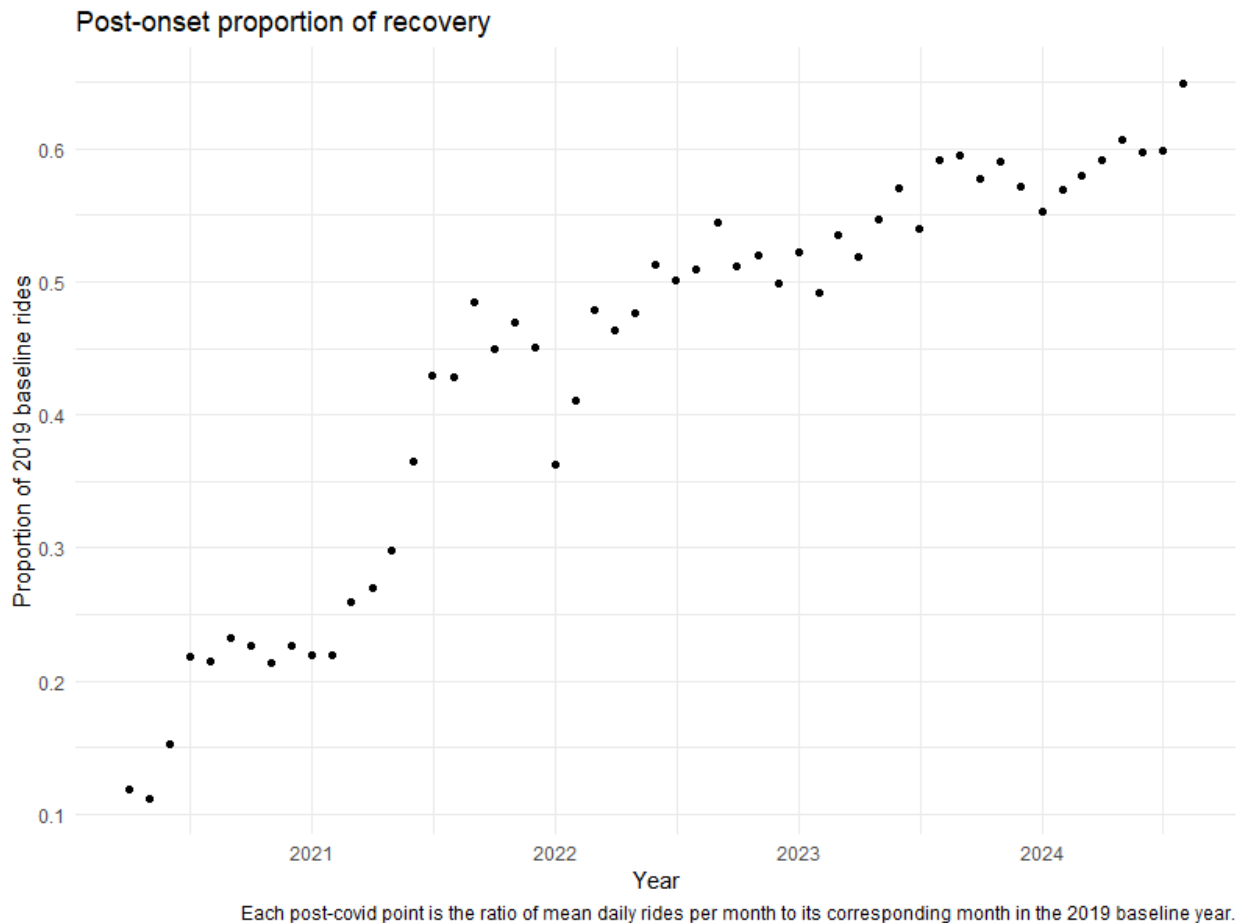


Figure 2: figure 02, Proportion of ridership recovery, following the onset of the COVID crisis.

Referring to figure 02, the magnitude of ridership recovery is clearer when proportion/percentage of recovery is being measured. There is an obvious incline in mean daily rides per month, with the last point at ~0.649

or about 65% recovery of ridership as of 2024-08-01. This indicates that ridership experienced a sustained recovery, but had not returned to pre-pandemic levels by August 2024, over a year after the official end of the COVID-19 global health crisis, announced in May 2023. Although there is a gradual incline, the rate of incline is not constant. Roughly, the fastest increase in ridership occurred between 2021 and 2022, and gradually slowed down thereafter.

Recovery and income trends

3 Is there variation in ridership recovery by income level of L station's location/area?

Yes, there are marginal differences in variation in ridership recovery by income level of each L station's area/neighborhood.

Q2: What were the variations in CTA L ridership recovery according to the economic characteristics (income) of the census tract that contains the CTA station?

Available data provides income “buckets” of which households are placed in for each Chicago neighborhood. To measure the income level of a Chicago neighborhood (and census tract), I construct a column, “income index,” that ranges from 1-5 (low to high) and is a *weighted* measure of the mean income bucket/bracket of the neighborhood and census tract of which contains the station. Each station record has a corresponding income index or “score.” It is then split into tertiles. The 33rd percentile is the upper limit of the “Low” income group, the 67th percentile the upper limit of the “Middle” income group, and the remaining portion is the upper limit of the “High” income group.

Figure 03 shows varied recovery by income group in the early years of the COVID crisis. Following the sharp cut-off in 2020, stations in the Low income group retained a substantially larger proportion of their 2019 ridership than stations in the Middle and High income groups. The plot visually indicates *correlation* between income group and ridership recovery.

The differences in recovery between income groups gradually narrowed into the later years of the pandemic. Stations in middle and high income areas experienced comparatively faster recovery from their baseline levels, and by late 2021 to 2022, recovery ratios across the three income groups had largely converged. From 2022 onward, all income groups followed similar upward trajectories. By 2024, all three income groups had recovered to ~65% of their 2019 baseline levels.

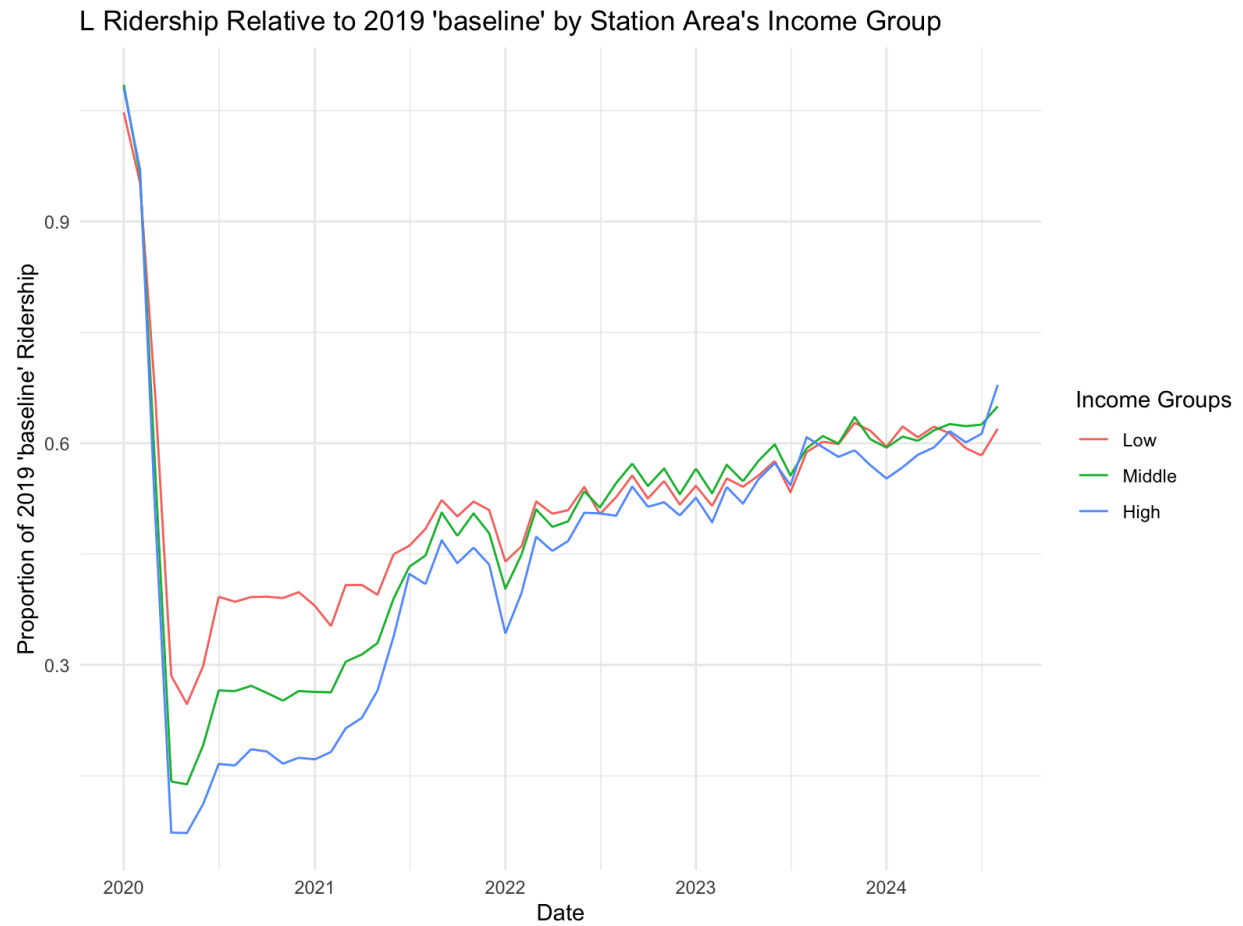


Figure 3: figure 03, Ridership recovery by each station area's income group.

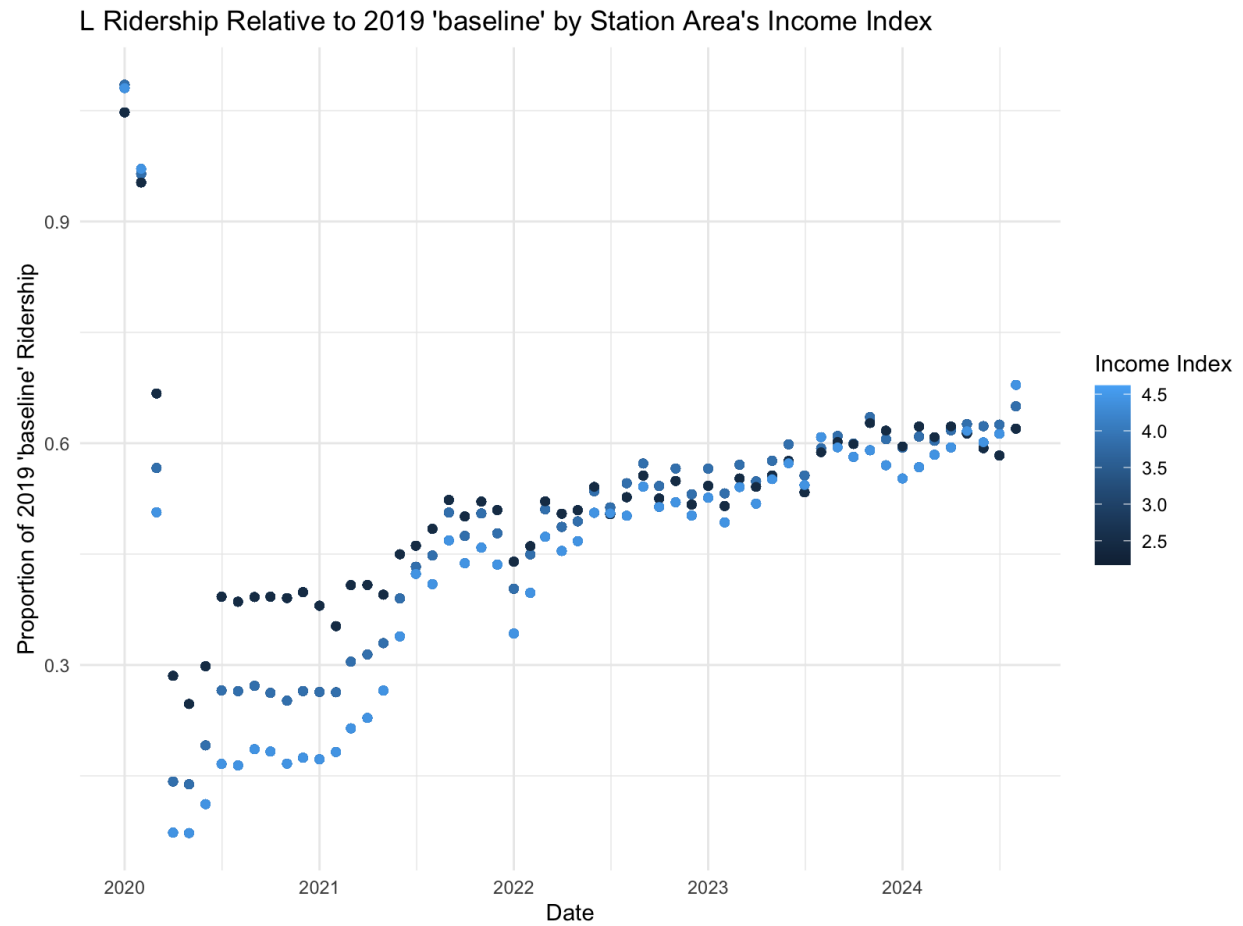


Figure 4: figure 04, Ridership recovery by each station area's income index.